Nick Marmaras

XR Software Engineer — Medical VR / R&D

CONTACT

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SUMMARY

XR Software Engineer and Technical Lead with 5+ years of experience delivering production medical VR applications and R&D projects. Strong in Unity, C#, Unreal Engine and C++, with hands-on experience in project coordination, developer mentoring, software architecture, real-time systems and performance optimisation. Experienced integrating AI capabilities and using AI-assisted development tools to improve engineering workflows, while maintaining code quality and reliability.

SKILLS

- **Leadership & Delivery:** Technical leadership, project planning, mentoring, task allocation, code reviews, releases
- **XR & Development:** Unity, C#, Unreal Engine, C++, OpenXR, Meta Quest, Photon PUN 2, hardware integration
- **AI & Automation:** OpenAI Codex, Claude, Unity MCP, prompt engineering, AI-assisted development, Azure AI
- **Engineering & Quality:** Git, CI/CD, testing, debugging, profiling, optimisation, documentation

PROFESSIONAL EXPERIENCE

XR Software Engineer (Unity/Unreal) **July 2021 - Present**
ORamaVR SA

- Lead technical delivery across multiple medical VR simulation projects, coordinating development priorities, milestones, implementation, testing and releases.
- Mentor and coordinate developers through task assignment, technical guidance, code reviews and constructive feedback, while remaining actively involved in development.
- Design, develop and maintain production Unity/C# and Unreal Engine/C++ systems within the MAGES SDK, including interactive training modules, analytics and multiplayer workflows.
- Use AI development tools, including OpenAI Codex, Claude and Unity MCP, for code exploration, prototyping, refactoring and debugging, validating generated changes through code review and testing.
- Develop and integrate AI-enabled functionality, including JARIA components and Azure AI Speech-to-Text, supporting real-time voice-driven simulation workflows.
- Prototype and integrate XR hardware, including Vive Trackers and Vive Ultimate Trackers, implementing real-time motion tracking, calibration and hardware-to-software interaction.
- Profile and optimise standalone VR applications, investigating CPU/GPU bottlenecks, memory constraints, frame-rate stability and runtime reliability.
- Collaborate with medical experts, researchers and engineering teams to translate complex requirements into production-ready software.

PROJECTS AND PUBLICATIONS

JARVRIKS — Full-Body VR Tracking & IK System

- Developed a multithreaded full-body VR IK system in Unity using Vive XR Elite and Vive Ultimate Trackers, including runtime calibration and real-time effector manipulation.
- Built upper- and lower-body interaction demonstrations to evaluate tracked movement, calibration and responsive full-body interaction.

VR Isle Academy — VR Digital Twin for Robotic Surgical Skill Development | CGI 2024

- Co-developed a device-agnostic medical VR digital twin using tracked hand and foot interaction with real-time/offline performance monitoring.
- Published in Advances in Computer Graphics, Springer LNCS.

EDUCATION

Doctoral Researcher — Hellenic Open University | Part-time alongside full-time employment (2024–Present)

- Doctoral research in multi-user and adaptive VR training systems, combining distributed software architectures with data-driven/AI-assisted methods for adaptive training, resource management and experimental evaluation.

University Of Western Macedonia, MS in Digital Games and Multimedia Applications (Mar 2024)

- Focused on software technology for games, XR, multimedia applications, C#, Python/Pygame prototyping, and advanced interactive systems. Built Python/Pygame prototypes including a 2D fighting game and a retro raycasting shooter inspired by Doom/Wolfenstein, strengthening UI, event-loop, collision, and real-time interaction.

University of Crete, BS in Computer Science (Sep 2022)

- Relevant coursework/projects: operating systems, systems-level programming, C/C++, Java, data structures and algorithms, mini-shell development, OS/kernel-related exercises, QEMU-based experimentation, digital design, computer organization, and RISC-V.